GENERAL SPECIFICATION G7.9

One-Way Voice Communication System

CONTENTS

G7.9 ONE-WAY VOICE COMMUNICATION SYSTEM

G7.9.1 General

(a) References

(i) The following is a list of standards that may be referenced in this Section:

1) International Electrotechnical Commission (IEC)

Standard	Subject
BS 6724	Specification for 600/1,000V and 1,900/3,300V Armoured Electric Cables Having Thermosetting Insulation and Low Emission of Smoke and Corrosive Gases When Affected by Fire
IEC 60332	Tests on Electric Cables under Fire Conditions
IEC 60754-1	Test on Gases Evolved during Combustion of Materials from Cables – Part 1: Determination of the Amount of Halogen Acid Gas
IEC 60754-2	Test on Gases Evolved During Combustion of Electric Cables – Part 2: Determination of Degree of Acidity of Gases Evolved During the Combustion of Materials Taken from Electric Cables by Measuring pH and Conductivity
IEC 60794	Optical Fibre Cables – Part 1-1: Generic Specification – General
IEC 60794	Optical Fibre Cables – Part1-2: Generic Specification – Basic Optical Cable Test Procedures
IEC 61034	Measurement of Smoke Density of Cables Burning Under Defined Conditions

2) SPRING Singapore

Standard	Subject
CP 5	Electrical Installations
SS 551	Code of Practice for Earthing
SS 546	Emergency voice communication system in buildings
SS 555	Lightning Protection
SS 299	Fire Resistant Cables

G7.9.2 Scope Of Work

(a) This Specification is to provide the operational concepts and the minimum performance and technical specifications for the equipment forming the integrated Public Address/One-Way Voice Communication System (PA/EVC) to be installed in the server room, electrical switch rooms and other buildings to be constructed under the contract. The Contractor is required to provide Paging/Emergency announcement facilities within the buildings which can integrate with the existing system, if applicable, without any implication on the system.

(b) All equipment and components used shall be of the same make and type for uniformity of standards. The work scope shall comprise the design, supply, delivery, installation, testing and commissioning.

(c) The Contractor shall furnish and install the complete system which is certified for compliance of IEC 60849 as specified with all necessary labour, apparatus, equipment, wiring etc., required to ensure on completion, a system of professional quality in excellent working order.

(d) Submittals

(i) Shop Drawings

1) As a minimum, submit the following items:

- Full equipment list and catalogues with equipment layout details and schematic diagram of the entire system;
- Schedule of technical data;
- The measurement data from manufacturer;
- Test equipment available; and
- A listing of key personnel available during the installation and defects liability periods along with their sound system qualifications and acoustics experience.

(ii) Quality Control Submittals

1) The following documents shall be submitted free of charge in a hard cover binding as part of the whole system:

- Complete set of instruction manuals for the system components;
- System schematic diagram;
- Service instructions;
- Copies of system operation instruction sheet laminated with hard clear plastic; and
- A statement, duly signed by the Contractor, for a guarantee of 2 year from the date of acceptance.

(iii) Samples

1) Submit samples of the following items for acceptance by the Superintending Officer prior to the Installation:

- Two 1 m samples of each type of cable to be used;
- Volume controls; and
- One sample of each type of speaker, etc.

G7.9.3 General Equipment Specification

(a) The following specify the technical requirements of all equipment to be supplied and installed under this Contract. The tenderer shall highlight any deviation from these Specifications in the submittal.

(b) The PA controller shall be based on 19" rack frame and shall comprise at least the following:

(i) Network Controller;

(ii) Power Amplifier with line supervision cards;

(iii) Display and Keyboard;

(iv) Power Supply; and

(v) Distribution System Software.

(c) The Network Controller shall communicate and address all function modules and interface circuitry. The Microprocessor shall continuously check and guard the system hardware for errors, malfunctions of disconnections. If a malfunction is detected, a message shall be displayed on the display.

(d) It shall comply with following minimum requirements:

(i) Public address and emergency control unit;

(ii) Fully digital;

(iii) Minimum 8 supervised control inputs and 5 control outputs;

(iv) Minimum 4 audio inputs and 4 audio outputs;

(v) Ethernet interface for configuration and diagnostic & logging functions;

(vi) Digital message storage. Stores up to last 200 fault messages;

(vii) The audio line outputs have a selectable 20 kHz monitoring signal;

(viii) The control outputs programmable for faults or calls and connected to visual and audible fault indicators;

(ix) The network controller able control up to 60 nodes of power amplifiers, call stations etc;

(x) The network controller is designed for redundant network cabling;

(xi) The network shall be wired as a redundant loop;

(xii) Shall handle 256 priorities, and can be configured for up to 244 zones;

(xiii) The unit shall have a switch mode power supply;

(xiv) The network controller is provided with a storage facility to store the last 200 fault messages in the system;

(xv) Automatic messaging is included in the network controller;

(xvi) Provide compact flash memory card, with memory size according to the storage requirements for audio messages. Four audio messages shall be able to be played simultaneously;

(xvii) The status of the digital audio message storage and of the messages themselves is monitored. Audio messages (as a set of wav files) can be downloaded from a computer via the Ethernet link;

(xviii) The network controller monitors the status of all the equipment in the system including call station microphone to the end of a loudspeaker line and reports status changes;

(xix) The external cables connected to the control inputs shall be monitored for short and open circuit;

(xx) Internally generated pilot tone shall be provided to the audio outputs for monitoring the outputs;

(xxi) Provision to store 7 attention tones, 3 test tones and 45 alarm tones, which can be accessed by any call stations or control inputs for announcement broadcast or alarm broadcast;

(xxii) The network controller has an internal real time clock for automatic playback of scheduled messages or automatic volume changes of announcements or background music;

(xxiii) Audio processing like equalization, limiter and gain for audio inputs and audio outputs shall be adjusted using the configuration software;

(xxiv) The network controller shall be provided with a monitoring loudspeaker and a headphone socket for monitoring of audio channels;

(xxv) Analog Line Inputs

1) Frequency Response:
   - -3 dB at 20 Hz and 20 kHz (tolerance ±1 dB); and
   - 0 dB at 100 Hz, 1 kHz and 10 kHz (tolerance ±1 dB).
2) Signal to Noise Ratio > 87 dB(A) at maximum level; and
3) CMRR > 40 Db.

(xxvi) Analog Microphone Inputs

1) Frequency Response:
   - -3 dB at 20 Hz and 20 kHz (tolerance ±1 dB); and
   - 0 dB at 100 Hz, 1 kHz and 10 kHz (tolerance ±1 dB).
2) Nominal Input Level -57 Db;
3) Signal to Noise Ratio > 62 dB(A) for dynamic range < 25 dB;
4) CMRR > 55 dB at 100 Hz;

5) Input Impedance 1360 ohm; and

6) Phantom Supply 12 V ±1 V @ 15 mA.

(xxvii) Audio Outputs

1) Frequency Response:

- -3 dB at 20 Hz and 20 kHz (tolerance ±1 dB); and
- 0 dB at 100 Hz, 1 kHz and 10 kHz (tolerance ±1 dB).

2) Output Impedance < 100 ohm;

3) Signal to Noise Ratio > 89 dB(A) at maximum level; and

4) Distortion at 1 kHz < 0.05%.

(xxviii) Temperature -5°C to +55°C;

(xxix) Humidity 15% to 90%;

(xxx) Safety According to IEC 60065; and

(xxxi) Immunity According to EN 55103-2.

G7.9.4 System Description and Functional Specifications

(a) The system installed shall designed for simple hook-up and shall be fully transistorised, allowance of adequate facilities for future expansion and servicing of equipment. The entire system shall be connected to UPS supply.

(b) The system shall be capable of fulfilling the following requirements:

(i) Clear, un-distorted announcements to selected areas during public addressing and emergency announcements;

(ii) Clear, un-distorted paging to all zones; either individually or collectively. Selection of groups of zones shall be programmable from time to time; and:

1) Flexibility to allow allocation of background music to selected areas when the other functions are not selected; and

2) It shall be possible to set different time settings for the background music to be on.

(iii) Central monitoring of all system status on a central existing system.

(c) The system shall be of digital technology and expandable as and when it is required. A controller shall be provided in this Contract with the possibility of linking up to the future controllers in the other building without having to shift the controller or interrupt the system operation.

(d) The controller shall be micro-processor controlled, and shall be flexible and easy to operate.

(e) Each of these controllers in each phase shall be linked up in the network via simple cabling such that it shall be possible to make announcements from any of the call stations in any building to any of the pre-assigned zones in the other building. Ultimately all the building will be operated as a complete integrated system. With this integration, paging can be done from any of the call stations to all/specific/selected group loudspeakers zone in all call mode /specific mode /selected mode from any of the call stations. Provision shall also include paging to all zones under all call mode within each building via the existing PABX/telephone sets (fixed and mobile). Provision shall also include interface with Fireman's Intercom (Two-Way Voice Communication System). All necessary provision within the controller shall be deemed to be included in the Contract.

(f) Each user key on the call station shall be programmable via the PA controller's built-in keyboard and display. Each key shall be capable of being programmed with a priority hierarchy, signal tones, digitally stored messages and routing instructions prior to a call. Each user key when depressed shall be able to activate a zone or group of zones of speakers.

(g) A central control system is to be provided for new system status monitoring; this includes monitoring of the control sub-system, power amplifiers, loudspeaker lines, UPS / battery charger, alarms and status, etc. It should log all paging call activities, including announcements of pre-recorded emergency messages. The system shall be capable of communicating to a host computer system over an Ethernet connection for configuration purposes and can be work independently without the computer connected to it. A local display of 2x16 character LCD display with menu button to navigate items and messages.

(h) The paging system shall be a one-way microprocessor based communication system of 12 zones which shall be fully expandable to 244 zones and beyond. Paging and announcements through the sound system shall be by simple activation of a switch. A selectable chime signal shall precede the paging and shall be initiated by the "paging switch".

(i) All loudspeaker zones can be selected with several zones combined or individually by selecting the respective user key.

(j) The PA controller shall have its built-in watch dog timer; all modules connected as well as all inputs connected, including the call stations shall be fully supervised and any fault detected shall be automatically reported on the control panel, and on the central monitoring system.

(k) The system shall be capable of multiple channel inputs so that more than one announcement can be made simultaneously.

(l) The central processor shall have a system of priorities such that, should a conflict situation arise, the station or user key with the top priority will override the others. This sequence of priorities shall be determined and programmed during the commissioning stage; it shall be possible to change the sequence by re-programming, as and when the need arises.

(m) The zones shall further be grouped according to function so that it shall be possible to make an announcement by depressing just one user key on the call station.

(n) Switching of the zones shall be done on the combination of input and output side, zones which are expected to receive many announcements should have its dedicated power amplifier, whilst other zones mainly for emergency announcements only shall have a combination of few zones sharing an amplifier.

(o) When the zones are selected for public addressing, a chime shall first be heard, followed by the announcement. The system shall have a range of chimes such that it shall be possible to program different chime types for different user keys and / or different call stations.

(p) It shall be possible for the system to function with different call stations in operation simultaneously, provided there is no conflict in the zones being called by the call stations.

(q) Where additional microphone points are required, they shall be terminated with a high quality XLR three pin connector and shall be connected to the amplifier via screened cables. Speeches and announcements can then be made over the public address network via a heavy-duty microphone plugged into one of these points.

(r) Speaker volume can be adjusted by zonal volume control (located at the equipment racks) or by local speaker volume control as specified. The local speaker volume control unit shall be a standard flush-mounted type with a control knob and each shall be installed beside the light switch location.

(s) All the equipment for the one-way voice communication system shall be designed for rack mounting, design of which shall be to the approval of the Superintending Officer. The rack shall be of the totally enclosed free-standing type. The rack shall be designed for easy maintenance and operation. It shall be equipped with forced ventilation fans. All equipment, switches, etc., shall be labelled. All equipment supplied shall be maintenance free, high quality, up-to-date and elegant in design. The design concept and layout shall be such that its operation shall be fail-safe and easy to operate.

(t) All equipment installed, e.g. microphone pre-amplifiers, line amplifiers, zone relay unit shall be of a modular type such that in the event of any failure of these units, they can easily be replaced by unplugging the faulty unit and plugging in a new unit.

(u) The system shall provide paging announcement facilities. The system shall minimally comprise the following:

(i) 28 audio channels microprocessor-based PA controller, further expandable;

(ii) Universal pre-amplifier suitable for microphone and line inputs; this shall have 4 inputs and can be set to either 4 outputs, or a common output. It shall also have bass and treble controls for each input;

(iii) 100V line power amplifiers with monitoring facilities;

(iv) Speaker line supervisory unit;

(v) Amplifier changeover monitor panel c/w amplifier monitoring;

(vi) The system shall include a stand-by amplifier for every 10 amplifiers;

(vii) Ceiling speakers, box speakers, horn speakers and column speakers as specified;

(viii) Paging call stations;

(ix) Volume controls;

(x) Facility for priority setting and control; this shall be a function included as standard in the controller;

(xi) Junction box c/w volume controls, where required;

(xii) Emergency message recorder for recording of emergency announcements from both the 1 way and 2 way fireman intercom system;

(xiii) Cassette/CD player, FM/AM tuner;

(xiv) Interface units with PABX/Phone set, Fireman Intercom System, Fire Alarm System, and MCS controller;

(xv) Standby power supply unit; and

(xvi) All necessary cables, steel conduit, steel trunking and surge arrestor.

(v) Emergency Call

(i) It shall enable a one-way announcement or a pre-recorded message to be given to all zones in an emergency situation. Emergency messages shall have priority over all other usage.

G7.9.5 Materials

(a) General

(i) All equipment supplied shall be capable of operating effectively and efficiently under the tropicalized site conditions.

(b) Materials

(i) All materials shall be of new and unused quality. All equipment and materials previously installed or used will be rejected. Materials and equipment shall be stored in such a manner as to be in a new condition when installed and to avoid damage from weather and site conditions. Damaged, deformed, and cracked equipment or materials will be rejected. Replacement shall be the responsibility of the Contractor at no additional cost to the Board.

(ii) Materials and equipment to be incorporated into the works (as called for in this Specification) are required to meet the quality/testing standards of designated institutes, societies, and standards associations. However, equivalent materials and equipment items meeting other authoritative standards which ensure an equal or higher quality than the standards mentioned may also be accepted if the Superintending Officer gives his approval.

(c) System Configuration

(i) The system shall have minimum 28 audio channels which can be expandable and minimum 3 call stations. All power amplifiers shall have back-up amplifier and each amplifier shall have spare. For each type of power amplifier, the same type of power amplifier has to be included as back-up.

(ii) The system shall be flexible to different power requirements of zones.

(iii) Amplifiers monitoring and change-over shall be incorporated in the amplifier itself.

(iv) Loudspeakers line monitoring shall be incorporated in the power amplifier by the supervision master card in the power amplifier and installing the supervision slave card at the end of the loudspeaker line. These cards also have to be installed in the back-up amplifiers.

(v) All the call stations shall be installed with extension keypads.

(vi) The existing call station at the SD building shall be re-programmed for the added zones under this contract. It can be replaced if the existing call station is not capable of zone extension and all the existing zones shall be included.

(vii) Minimum two zones shall be provided with local audio inputs. These inputs shall be connected to the audio inputs of the corresponding power amplifiers, and can be given a lower priority than any announcements and can be overridden if there is an announcement with a higher priority.

(viii) The network controller shall have minimum 4 audio inputs. The network controller shall be configured with redundant communication interface either by fibre optic if the distance between the call station nodes are more than 100 meter or by copper cables.

(ix) The configuration software is used to configure the system.

(x) The diagnostic and logging software shall be provided for monitoring of the various system units and display of status changes on the computer.

(xi) Central Equipment Rack

1) Provide a standard '19 inch' rack or rack to house the equipment. The word 'rack' as used throughout this Specification refers to a single rack or a system of multiple racks, as necessary.

2) Install equipment within the rack in a logical (signal path) sequence. Notwithstanding this requirement, locate equipment requiring routine operation (e.g. switch and monitor panels) at a convenient working height. Equipment requiring internal access during commissioning and/or routine testing shall preferably be installed on rack runners. The equipment shall remain fully operational when extended on the runners.

3) Items requiring routine operation shall, where practicable, be installed within the same cabinet.

4) Fit unused rack space with blank panels.

5) Provide an internal mains power (230V) distribution system. Label each termination with the equipment item served.

6) Shield all mains terminals from inadvertent touch.

7) Provide a mains power panel with an indicator light at the bottom of the rack. Provide standard 13A or IEC socket for test equipment. This shall be independent of the rack power switch and shall be annotated accordingly, i.e. 'UNSWITCHED'.

8) Provide contact details for the equipment supplier (name, address, telephone and fax numbers) on a permanent nameplate affixed in a prominent position on the rack.

9) Where specified provide an integral drawer for storage of spares, manuals, etc and an integral pull-out shelf as a working surface for test equipment.

(xii) Power Amplifiers

1) Provide compact single output or modular power amplifiers. The amplifiers shall be sized to provide the specified loudspeaker output and area sound levels, with adequate headroom, allowing for appropriate diversity of loudspeaker tappings and loudspeaker line losses.

2) The total output power shall be rated to drive the entire system loudspeakers at a fully rated loading.

3) The system shall include minimally a stand-by amplifier for every 10 amplifiers of the same rating.

4) The power amplifiers shall be driven at 80% capacity, with 20% as spare for possible future additional loudspeakers.

5) An amplifier monitoring system shall continuously monitor the functioning of the power amplifiers and shall automatically switch in a spare power amplifier in case of failure of any of the amplifiers.

6) Where technology is readily available, features of digital to analogue output and built in amplifier monitoring and line supervision should be included.

7) The amplifiers shall be fully protected against short and open circuits and shall include protection against damage to loudspeakers. Fuses (preferably front panel mounted) shall be clearly labelled with rating and type.

8) Each amplifier shall be clearly labelled with the loudspeaker zone(s) served. (For parallel bank amplification, the labelling shall indicate the amplifier number).

9) At least 2 amplifiers, 1 connected to each of 2 independent loudspeaker lines, shall serve each zone. This shall not preclude the use of parallel banks of amplifiers, provided that all the requirements of this Specification can be met with parallel bank amplification.

10) The power amplifier shall be designed and constructed to bring reliable power to PA applications. The versatile units shall be operated at peak efficiency no matter how continuous the operations are, and with no danger of short circuits or overloads. The unit shall come with protective circuitry that can withstand accidents and mismatched output connections that may occur during installation or use.

11) The power amplifier shall have the following technical specifications:

- Automatic volume control;
- Redundant network connection;
- Power ‘ON’ indication;
- Digital audio processing for each amplifier channel;
- Audio delay for each amplifier channel;
- 2 x 16-character display for monitoring Headphone connection and VU meter for audio monitoring;

- Minimum 8 control inputs and 8 control outputs;
- Minimum 4 audio inputs;
- Minimum 4 audio outputs with selectable 100 V/70 V/50 V outputs;
- Supervision of external cable connected to control inputs;
- Amplifier monitoring and back-up amplifier switching;
- Loudspeaker line and loudspeaker supervision;
- The power amplifiers receive input signals via the network;
- The control inputs are freely programmable for system actions, and priorities can be assigned to these inputs;
- The control outputs are freely programmable for faults and calls;
- The control input can be programmed using the configuration software;
- Each control input has to monitor the cable between the control input and the external switch for open circuits and short-circuits;
- The power amplifiers shall be connected in redundant loop;
- The unit shall be monitored continuously and status changes shall be reported to the network controller for fault/status reporting;
- The change-over with back-up power amplifier;
- Amplifiers shall have a 48 V DC back-up supply input;
- Analog line inputs;
- Frequency Response -3 dB at 50 Hz and 20 kHz (tolerance ±1 dB);
    - Signal to Noise Ratio > 87 dB (rms unweighted);
    - CMRR > 40 dB at 1 kHz;
    - Input Range -6 dBV to 6 dBV; and
    - Input Impedance 22 kohm.
- Analog Microphone Inputs:
    - Frequency Response -3 dB at 100 Hz and 16 kHz;
    - Nominal Input Level -57 dBV;
    - Signal to Noise Ratio > 62 dB(A);
    - CMRR 40 dB at 1kHz;
    - Input Impedance 1360 ohm;
    - Phantom Supply 12 V ±1 V @ 15 mA; and
    - Input Range -65 dBV to -50 dBV.

- Audio Outputs:
    - Frequency Response 60 Hz to 19 kHz (-3 dB);
    - Signal to Noise Ratio > 85 dB with pilot tone off;
    - Crosstalk between Amplifiers <-80 dB at nominal load for 1 kHz; and
    - Total Harmonic Distortion <0.3% (at 1 kHz) at 50% of rated output power.
- Mounted in a 19" rack;
- Temperature -5°C to +55°C;
- Humidity 15% to 90%;
- Emissions According to EN 55103;
- Immunity According to EN 55103-2; and
- Safety According to IEC 60065-98.

(xiii) Loudspeakers

1) The different types of loudspeakers shall be used for different acoustic environment and application.
    - Ceiling Speaker: All areas with false ceilings at 2.4m;
    - Wall Speaker: All service areas without false ceiling, and staircases at 2.4m;
    - Horn Speaker: Basements and plantrooms at 2.4m; and
    - Column Speaker: High bay areas, galleries and where shown at 2.4m.

2) The nominal sound level for announcements shall be at least 10 dB above the ambient noise level in the area or zone for paging and speech level. The system shall work on 100V line.

Area	Ambient Noise Level (dB)
General Public Area	70
Exhibition Area	65
Offices	60
Toilet	50
Staircase/galleries	50
Plantroom	90

3) The distance between any two ceiling speakers in the office areas shall not exceed 5 m. All speaker lines shall be supervised for open circuit fault, short circuit fault, and short to ground fault. Upon detection, the status of the fault shall be indicated by LED and a buzzer shall sound.

4) For zones provided with horn speakers, the frequency response shall be at least 350 Hz to 8000 Hz, whilst that for the sound projectors, shall be from 120-Hz to 15-kHz; maximum variation shall be in the range of plus/ minus 3 dB.

(xiv) Recessed / Surface Ceiling Speaker

1) All ceiling speakers shall be delivered with matching transformer and baffle grille. Contractor is to ensure the Wattage Tapping of the speaker shall be verified on site and ensured that the output sound level is sufficient for the respective areas.

2) Recessed/surface ceiling speakers shall have the following technical specifications:

- Rated Voltage: 100V;
- Rated Impedance: 10 kΩ; 3.3 kΩ; 2 kΩ;
- Power Handling Capacity: 6 W;
- Effective Frequency Range: 70- to 18,000-Hz;
- Opening Angle (4-kHz, -6 dB): 70 degrees;
- Sound Pressure Level at 1-kHz, 1 W, 1 m: 92 dB (A);
- Maximum Sound Pressure Level: 96 dB(A);
- Speaker Diameter: >180 mm;
- The speaker baffle shall be with spring mounting; there shall be no visible screws on the baffle; and
- Safety Standard: IEC 60065.

3) Where high ceiling loudspeakers are required, they shall have the following technical specifications:

- System Type: 2-way, full range;
- Frequency Response: 60-Hz – 20-kHz;
- SPL (1 Watt/1 Meter): 95 Db;
- Power Handling: 36 W; and
- Peak: 50 W.

(xv) Wall (surface mount wooden box) speaker shall have the following technical specifications:

1) Rated Voltage: 100 volts;

2) Rated Impedance: 1,667 ohms;

3) Power Handling Capacity: 6 W;

4) Effective Frequency Range: 200- to 15,000-Hz;

5) Opening Angle (4-kHz): 95 degrees;

6) Sound Pressure Level at 1-kHz, 1 W, 1 m: 90 dB(A);

7) Maximum Sound Pressure Level: 98 dB(A); and

8) Safety Standard: According to IEC 60065.

(xvi) Horn loudspeaker shall have the following technical specifications:

1) Rated Voltage: 100 volts;

2) Rated Impedance: 1,000 ohms;

3) Power Handling Capacity: 10 W;

4) Effective Frequency Range: 410- to 8,000-Hz;

5) Opening Angle:

- Horizontal: 38 degrees; and
- Vertical: 48.

6) Sound Pressure Level at 1-kHz, 1 W, 1 m: 104 dB(A);

7) Maximum Sound Pressure Level: 114 dB(A);

8) Safety Standard: According to IEC 60065; and

9) Water Protection: According to IEC 60529 IP 65.

(xvii) Column loudspeakers shall have the following technical specifications:

1) Rated Voltage: 100 volts;

2) Rated Impedance: 833 / 417 ohms;

3) Power Handling Capacity: 12/ 24 W;

4) Effective Frequency Range: 200- to 20,000-Hz;

5) Opening Angle at 1KHz:

- Horizontal: 150 degrees; and
- Vertical: 35 degrees.

6) Sound Pressure Level at 1kHz, 1 W, 1 m: 90 dB(A);

7) Maximum Sound Pressure Level: 103 dB(A); and

8) Safety Standard: According to IEC 60065.

(xviii) Loudspeaker volume controls

1) The volume control shall be the auto-transformer type and shall be provided with graduated scale on faceplate, back box, and incorporated with override relays.

2) Volume control units shall have the following technical specifications:

- Type: 19-inch rack assembly with six volume controls one 2HE panel;
- Power Handling Capacity: 100 W;

- Rated Voltage: 100V;
- Relay: 12-volt dc relay for override function;
- Total Impedance: 20 dB (100 W);
- Attenuation per Step: 2 dB;
- No. of Positions: 10; and
- Safety Standard: IEC 60065.

(xix) Junction Box c/w Volume Control:

1) Metal junction box c/w volume control shall be provided for each individual zone and/or as specified. The junction box shall be designed to perform with the following functions:
   - Able to select area with paging;
   - Able to perform volume control for each individual area; and
   - Able to disconnect for easy maintenance and fault finding.

(xx) Microphone

1) The paging microphone shall be the uni-directional electrets condenser type with a 24 cm black flexible gooseneck.
2) The standard call station shall comprise a table stand fitted with a high performance electric condenser microphone mounted on a flexible stem for easy adjustability. It shall include two LED's, for 'engaged' and 'ready-to-talk' indication, and 10 push-button user-keys.
3) Each call station shall have a built-in amplifier for line level output, plus a compressor / limiter in order to maintain signal strength regardless of changes in the speaking distance from the microphone.
4) Each user key on the call station shall be programmable via the distribution centre's built-in keyboard and display. Each key shall be capable of being programmed with a priority hierarchy, signal tones, digitally stored messages and routing instructions prior to a call. Each user key when depressed shall activate a zone or group of zones of speakers according to type of announcement to be made.

(xxi) Paging Microphone Call Station

1) Redundant network connection;
2) Power ‘ON’ indication;
3) Status/fault indications;
4) Differentiation in indications for higher- and lower-priority announcements;
5) Supervision of microphone capsule;
6) Functional and practical;

7) The call station shall have a limiter and a speech filter with a cut-off frequency at 340 Hz to improve intelligibility and prevents clipping of the audio input on low-frequency signals;

8) The call station can be connected to a maximum of 16 call station keypad units via a serial communication link;

9) The power supply to the call station keypad units is provided from the call station;

10) Should have a volume control for the monitoring loudspeaker and headset;

11) The call station can be programmed for momentary actions on make contact and toggle actions without repeat on make contact;

12) 256 priorities can be assigned;

13) Analog to digital audio conversion is handled by the call station;

14) The call station basic shall have a DSP which can be used for audio processing, including adjustment of sensitivity, limiter and parametric equalizer settings;

15) The monitoring loudspeaker is switched on when the call station activates a chime or pre-recorded message;

16) Shall have extension keypad attached; and

17) Audio Characteristics

- Nominal acoustic input level 75 to 90 dB SPL;
- SNR > 60 dB at 85 dB SPL; and
- Frequency Response 340 Hz to 13.5 kHz (-3 dB).

(xxii) Message Announcement System

1) The storage of the pre-recorded emergency messages shall be digital, provided by EPROM memory chips on modular card, which shall be a standard card within the PA controller.

(xxiii) Emergency Voice Communication (Evc) System

1) Shall comply with requirements of SS 546 and shall be integrated with the One-way voice communication system.

(xxiv) Emergency Override Unit

1) When the ‘emergency’ all call user key at the FCC is selected, all other ongoing signals shall automatically be over-ridden to allow the emergency messages to be announced;

2) Provide the following messages unless indicated otherwise:

- 'ALERT' (uncoded);
- Coded staff warning;
- 'EVACUATE';
- Staff warning cancellation;

- System test;
- The 'ALERT' message shall be 5s of an approved attention-drawing alert signal, followed by the words: 'ATTENTION PLEASE. ATTENTION PLEASE. WE ARE INVESTIGATING AN EMERGENCY CONDITION. PLEASE LISTEN FOR FURTHER ANNOUNCEMENTS.' The complete message (signal and words) shall be continuously repeated, with intervening time intervals of 30s;
- The coded staff warning message shall be 5s of an approved attention-drawing signal followed by the words: 'ATTENTION PLEASE. STAFF CALL 100'. The warning signal and words shall be immediately repeated once, and then with intervening time intervals of 30s;
- The 'EVACUATE' message shall be 5s of an approved attention-drawing evacuate signal, followed by the words: 'ATTENTION PLEASE. ATTENTION PLEASE. THIS IS AN EMERGENCY. PLEASE LEAVE THE BUILDING BY THE NEAREST AVAILABLE EXIT. DO NOT USE THE LIFTS.' The complete message (signal and words) shall be immediately and continuously repeated with no intervening time intervals;
- The staff warning cancellation message shall be: 'ATTENTION PLEASE. CANCEL STAFF CALL 100'. The message shall be immediately repeated once; and
- The test message shall be: 'PUBLIC ADDRESS SYSTEM TEST. THIS IS A TEST OF THE PUBLIC ADDRESS SYSTEM. THIS IS A FUNCTIONAL TEST FOR SYSTEM OPERATION, LOUDSPEAKER COVERAGE, SIGNAL LEVEL AND SPEECH INTELLIGIBILITY.'

3) Messages shall not be 'composed' from a number of stores;
4) Messages shall use an unaccented voice, or one with a light local accent;
5) Submit details of standard messages for approval;
6) Prior to installation of the equipment, provide, on a standard compact cassette or CD, a copy of each recorded message for approval;
7) Messages shall start from the beginning within 0.25s of activation;
8) There shall be no user controls on the unit;
9) Provide a message run indicator light or LED (preferably green) for each message channel;
10) Erasure of any message shall not be possible without the connection of external components;
11) The message unit shall provide the following minimum performance:
    - frequency response 150Hz - 5.5kHz ±2dB;
    - S/N >60dB at rated output; and
    - distortion <1% @ 1kHz at rated output.
12) The unit shall have automatic self-surveillance.

G7.9.6 Wiring and Transmission

(a) Provide cables and wiring as specified and assign a unique reference number to each cable. Identify all cables clearly and permanently with a label at each end.

(b) Ensure all cores are colour coded or numbered.

(c) Size loudspeaker cables to ensure not more than 0.5dB power loss, assuming an additional future power load of at least +10% on each circuit.

(d) When there is a fault in any part of the system, it shall automatically be reported on the central monitoring system. The type of fault shall also be displayed, e.g. amplifier No failure, etc.

(e) The transmission channel via the wires shall have sufficient bandwidth to satisfy the requirement of the whole system. The bandwidth of the whole system shall be better than 500-Hz to 8-kHz.

(f) Fire-resistant cables complying with SS 299(Cat C, W and Z) shall be used. Cables shall be of LSHF or LS0H and meet applicable requirements of IEC 61034 or BS 6724 for smoke emissions and IEC 60754-1 for acid gas emission and IEC 60754-2 for determination of degree of acidity. Cables shall also meet flame propagation test requirements of IEC 60332 Parts 1 and 3A. Where cable is exposed directly to the mounting surface, armoured protection is required.

(g) Loudspeaker lines for all distributed system shall be connected via a barrier strip terminal block housed in a distribution panel for easy testing and maintenance.

(h) Cabling between a call station and the central system shall merely comprise a two-core screened microphone cable for every 10 user keys.

(i) All fibre optic cable/converter and accessories shall meet requirements as specified in Section 16121.

(j) All audio equipment grounds in the sound control equipment rack shall be connected to a common point on the rack and both the common point and the rack shall be grounded as specified in SS 551, Earthing. No connection to the telephone service ground or the electric power ground is allowed as these are the potential sources of interference from one of the devices connected to these services.

(k) All audio lines, including microphone, line level, and loudspeaker lines shall be floating with respect to ground, neither side of the audio lines shall be grounded. No equipment employing single ended input or output shall be used in the system. If equipment has a single ended input, it must be provided with an input transformer to provide for floating condition. There shall be no electromagnetic and electrostatic hum in the system.

(l) All equipment shall be suitable for operation for 230V AC, 50-Hz power source with regulation at 7-10V of nominal voltage. All input and output components shall be properly matched.

G7.9.7 Connections and terminations

(a) Limit the numbers of in-line connections to the absolute minimum practicable. Make any connections within suitable steel junction boxes, permanently annotated 'PA SYSTEM'.

(b) Make connections to equipment racks via marshalling boxes, also annotated 'PA SYSTEM'. Fit the boxes with fixed terminal strips permanently annotated with unique circuit reference numbers.

(c) Make final connections between the marshalling boxes and the racks via suitable flexible cables of sufficient length to gain working access to the rear of the racks.

(d) Make internal rack connections either hard-wired or via professional type audio connectors, preferably of the latching type.

(e) Protection shall be provided by means of surge suppressors to any equipment including cables which may be vulnerable to voltage transients, in a manner approved by the S.O.. This shall include the incoming mains connections to the power supply unit. The equipment manufacturers' instructions shall be observed in applying the protection.

(f) Segregate PA cables from all other cables by at least 300mm in any direction as far as practical. Cables shall only cross at right angles.

G7.9.8 Power Supply

(a) The main power supply provided for use of the Public Address System shall be:

  (i) Voltage: 230V plus or minus 6 percent;

  (ii) Ampere: 15-amp; and

  (iii) Frequency: 50-Hz plus or minus 4 percent.

(b) The Contractor shall make provision for all necessary power supply units, voltage regulators, etc., to ensure that the equipment will perform satisfactorily.

(c) All necessary power supply required for the operation of the sound equipment after the main power point shall be supplied and installed by the Contractor.

(d) Standby Power Supply

  (i) The standby power supply system shall consist of maintenance free sealed lead acid battery bank, automatic charger, solid state inverter and automatic changeover switch and of capacity for at least 2 hours of continuous operation. The battery bank and the charger shall be house in a floor standing cabinet.

  (ii) The trouble indicator panel shall be housed on the door and consist of voltmeter, ammeter, power switch, charger switch (high rate/float rate, automatic) and corresponding high, float rate and automatic charging indicating lamps.

  (iii) The stand-by battery charger system shall have provision to send the alarm outputs to the central monitoring system.

(e) The Distribution System Software can be installed at any of the workstations and the display at the PA rack shall be connected with the work station.

(f) Surge Protection Of Public Address System

  (i) The system provided shall be protected against lightning induced over voltages in accordance with SS 555.

  (ii) All metal frames of the console shall be effectively earthed and the 230V single-phase power supply shall be protected by surge arrestors.

G7.9.9 Source Quality Control

(a) General

(i) Test all One-Way Voice Communication System equipment, including all components in accordance with IEC and relevant regulatory standard.

(ii) Testing to be in accordance with established ISO 9001 procedures.

(b) Factory Inspections

(i) Inspect for required construction, electrical connection, and intended function. All Source Quality Control and Factory Acceptance Tests (FAT) shall be witnessed by S.O. / the Board.

(ii) Notify S.O. at least 2 weeks prior to any testing to allow S.O. to witness tests.

(iii) Completely assemble the one-way voice communication system at the factory, and perform all Routine Tests at manufacturer's shop in accordance with IEC.

G7.9.10 Execution

(a) Installation Requirements

(i) The Work herein specified shall be performed by fully competent workmen in a thorough manner. The Contractor shall furnish all equipment material, cables and supplies to effect the complete installation of this system.

(ii) All materials furnished shall be new and all work shall be completed to the satisfaction of the Superintending Officer.

(iii) All equipment, except portable equipment shall be held firmly in place. This shall include loudspeakers, cables, etc. Fastenings and supports shall be adequate to support their loads with a safety factor of at least 3.

(iv) The Contractor shall be fully responsible in ensuring all supports to be structurally sound, especially those for the loudspeakers. All switches, outlets, etc. shall be clearly, logically and permanently marked during installation.

(v) Care shall be taken in wiring so as to avoid damage to the cables and to the equipment. All joint and connectors shall be made with resin cone solder or with mechanical connectors approved by the major equipment manufacturer.

(vi) All wiring shall be executed in strict compliance with the local codes applicable and with the standard broadcast practices. Cables for microphone level circuits (below levels minus 20 dBm), line level circuits (up to 30 dBm), loudspeaker circuits (above plus 30 dBm) and power circuits shall all be run in separate conduits.

(vii) The Contractor must take such precautions as are necessary to guard against electromagnetic and electrostatic hum, to supply adequate ventilation and to install the equipment so as to provide maximum safety to the operator.

(viii) In-situ soldering shall be kept to a minimum. Cables shall, as far as possible, be hooked up to equipment components via appropriate ring type cable lugs and screws or insulation displacement type terminal blocks. Where component leads/wires need to be cropped the electrically and mechanically. A wrap of turn round a terminal is considered minimum. Sufficient slack shall be allowed at connections to compensate for temperature changes and vibration. No wire shall be taut.

(ix) Bends for cables at the equipment connections shall have on inside radius no less than two times the cable diameter.

(x) Groups of cables shall be bunched using approved cable ties or straps to give a neat appearance. Except for twisted pairs, cables shall not be entwined.

(xi) Wherever conduit for wiring is to be concealed in the plasterwork of walls or buried in floors, they shall be installed in chases of sufficient depth and width to suit the sizes of conduits and parallel runs involved.

(xii) Where required, the Contractor shall cut the necessary chases in the wall for the installation of the conduits. Initial plastering of chases shall be undertaken by the Contractor, after conduits have been erected in position.

(xiii) Where conduit is to be cast in concrete at the site, the Contractor is required to communicate very closely with the Civil Contractor and arrange for all the conduits to be laid in position before the casting of any concrete is carried out by the Civil Contractor throughout the progress of the construction.

(xiv) The methods employed for the installation of concealed conduit wiring shall generally be the same as that for surface-run conduit wiring except that conduits shall be held in their respective chases by means of normal, galvanised metal saddles and no painting of conduits before plastering or cementing is required, except where such conduits rise from floors or emerge from walls.

(xv) Where it is not possible for rigid conduit to terminate at the equipment, then such conduit shall be connected to approve flexible conduit with special adapters and the flexible conduit, in turn, shall terminate at the equipment. Flexible conduits used throughout shall be of the liquid-tight.

(xvi) All cables from the equipment rack to the distributed speakers shall be carried out using at least 1.5 mm^2 cable, meeting all applicable requirements of SS CP25.

(xvii) All precautions shall be taken by the sound system Contractor to prevent cross talk and spurious oscillations. There shall be no electromagnetic and electrostatic hums and noise in the system.

(xviii) The system shall be guaranteed against materials defects, design, workmanship and improper adjustment.

(xix) All equipment, except portable equipment, shall be held firmly in place. This shall include speakers, amplifiers, racks, cables, etc. Fastenings and support shall be adequate for support of their loads with a safety factor of at least 3. All switches, connectors, outlets, etc., shall be clearly, logically and permanently marked during installation.

(xx) All components utilised shall be professional grades of known and reputed manufacturers.

(xxi) Wherever it is found that several surface-run conduit cable runs are grouped together, thereby utilising extensive space on walls or steel structural work, the Contractor may, with the Superintending Officer's approval, use steel trunking in lieu of conduit for cable runs.

(xxii) The trunking shall be manufactured from zinc-coated, mild sheet steel of 1.6 mm thickness for cross-section sizes of up to and including 300 mm by 100 mm. For sizes exceeding 300 mm by 100 mm, 2 mm sheet steel shall be used.

(xxiii) Steel trunking may be supplied in lengths of 2 m or 4 m, with each length being provided with a sleeve-type coupling and external earth bonding link of copper. All trunking shall have a smooth interior with cover plates overlapping the sides of the trunking.

(xxiv) In general, the trunking shall be designed for exceptional strength and rigidity with trunking lengths fitted with "butt-up" joints to form the necessary runs. Suitable adapters shall be utilised for any change in a cross-section of trunking runs. All such adapters, bends, tee pieces and stop bends used in the trunking installation shall be fabricated from the same material as the trunking and fitted with sleeve type couplings at each trunking connection point.

(xxv) For every length of steel trunking run, a copper tape earth continuity conductor (25 mm by 3 mm cross-section) shall be provided throughout the full length of the trunking run and bonded to the trunking at every section.

(xxvi) In vertical trunking runs, insulated type cable support pins and retaining clips shall be fitted to support the weights of the cables.

(xxvii) Where conduit is tapped off trunking, suitable brass, smooth bore bushes shall be fitted at all conduit terminations. All-insulated type, plastic fibre, bake or Bakelite bushes for this purpose shall not be used.

(xxviii) All trunking is to be finished with two coats of an approved-type light green paint.

(xxix) The microphone level conduits, line level circuits, loudspeaker circuits and power circuits shall run in separate conduits/trunking. All other conduits shall be well spaced from power conduits. Power conduits shall be earthed with heavy shielded wire to the power system earth. Low voltage dc circuits for relay control could be run in any conduit except microphone line. Microphone and 600 ohms line shall be insulated from the conduit and from each other for the entire conduit length. Microphone lines and 600 ohms line entering and leaving the equipment racks shall be connected via moulded terminal blocks.

G7.9.11 System Testing

(a) The Contractor shall test the system in the presence of the Superintending Officer to show that its performance satisfies the requirement of this Specification.

(b) All test equipment shall be professional and supplied by the Contractor. A sound pressure meter will be required.

(i) Sound level meter and calibrator, complying with BS 5969 (IEC 651) Type 1 or 2;

(ii) ⅓ octave band spectrum analyzer (minimum range 40Hz -16kHz at ISO centre frequencies, display resolution increment not more than 2dB);

(iii) Pink noise source (minimum bandwidth 20Hz - 20kHz, ±0.5dB) with an amplifier for and sufficient loudspeakers to produce a noise level within 5dB of the overall area signal level(s) for each area containing an ambient noise sensing transducer. The noise level shall be uniform throughout the area to ±2dB of the mean, measured 1.5m above the floor;

(iv) Oscilloscope for waveform analysis;

(v) Field strength meter for audio loop;

(vi) Test headset for audio loop analysis;

(vii) Handheld transceivers approved for use on site; and

(viii) Provide current calibration certificates for all the above equipment.

(c) If required by the SO/Engineer (in the event of apparent non-compliance of the installed system with the Specification), then provide precision test equipment for verification of voltage, impedance, power, frequency response, phase response, signal-to-noise ratio, total harmonic distortion and RASTI.

(d) The cost of this test shall be deemed to be included in the Tender sum.

(e) Manufacturer’s Field Services

(i) Manufacturer’s Representative

1) Present at site or classroom designated by S.O. for minimum person-days listed below, travel time excluded:

- 2 person-days for installation assistance and inspection;
- 2 person-days for functional and performance testing and completion of Manufacturer’s Certificate of Proper Installation;
- 1 person-day for pre-start-up classroom or site training;
- 1 person-day for facility start-up;
- 3 person-day for post-start-up training of the Board’s personnel.

2) Training shall not commence until an accepted detailed lesson plan for each training activity has been reviewed by S.O..

END OF SECTION